Calculus 12 - Transcendental Functions & Integrals RA (V1)

Name : _____

Teacher's name: _____

Outcome	0	1	2	3	4	5	6	7
Reasoning and Analyzing								

1. For each of the following statements, identify if the statement is **always true**, **sometimes true**, or **never true**. Provide a brief justification.

   (a) $y = \cos x$ is concave up whenever $\cos x > 0$.

   (b) If $f(x) = e^{ax}$ where $a \neq 0$, the function is decreasing everywhere.

   (c) If $\int_a^b f(x)\,dx = m$ and $\int_b^c f(x)\,dx = n$, then $\int_c^a f(x)\,dx = -m - n$.

   (d) If $\dfrac{dy}{dx} = \dfrac{1}{x}$ where $x \neq 0$, then $y = \ln x + C$.

   (e) If $f(x)$ is a positive function and $f''(x) < 0$ everywhere, then $LRAM_4$ on any interval will be an over-approximation for the area beneath $f(x)$.

2. A student was asked to evaluate the definite integral $\displaystyle\int_1^e \frac{(\ln x)^2}{x}\,dx$ using the substitution $u = \ln x$. Which step did he **first** make an error? Explain the error.

$$\int_1^e \frac{(\ln x)^2}{x}\,dx = \int_1^e u^2\,du \qquad \textbf{Step 1}$$

$$= \left[\frac{u^3}{3}\right]_1^e \qquad \textbf{Step 2}$$

$$= \frac{e^3}{3} - \frac{1^3}{3} \qquad \textbf{Step 3}$$

$$= \frac{e^3 - 1}{3} \qquad \textbf{Step 4}$$

3. In each of the following, one item doesn't belong.

   (a) Which integral doesn't belong? Explain your reasoning.

   i. $\displaystyle\int_0^{2\pi} \sin x\,dx$ ii. $\displaystyle\int_{-\pi/2}^{\pi/2} \sin x\,dx$ iii. $\displaystyle\int_0^{\pi} \sin x\,dx$ iv. $\displaystyle\int_{-1}^{1} x\,dx$

   (b) Which function doesn't belong based on its derivative? Explain.

   i. $y = e^{3x}$ ii. $y = 3\sin x$ iii. $y = \tan(3x)$ iv. $y = \cos(3x)$